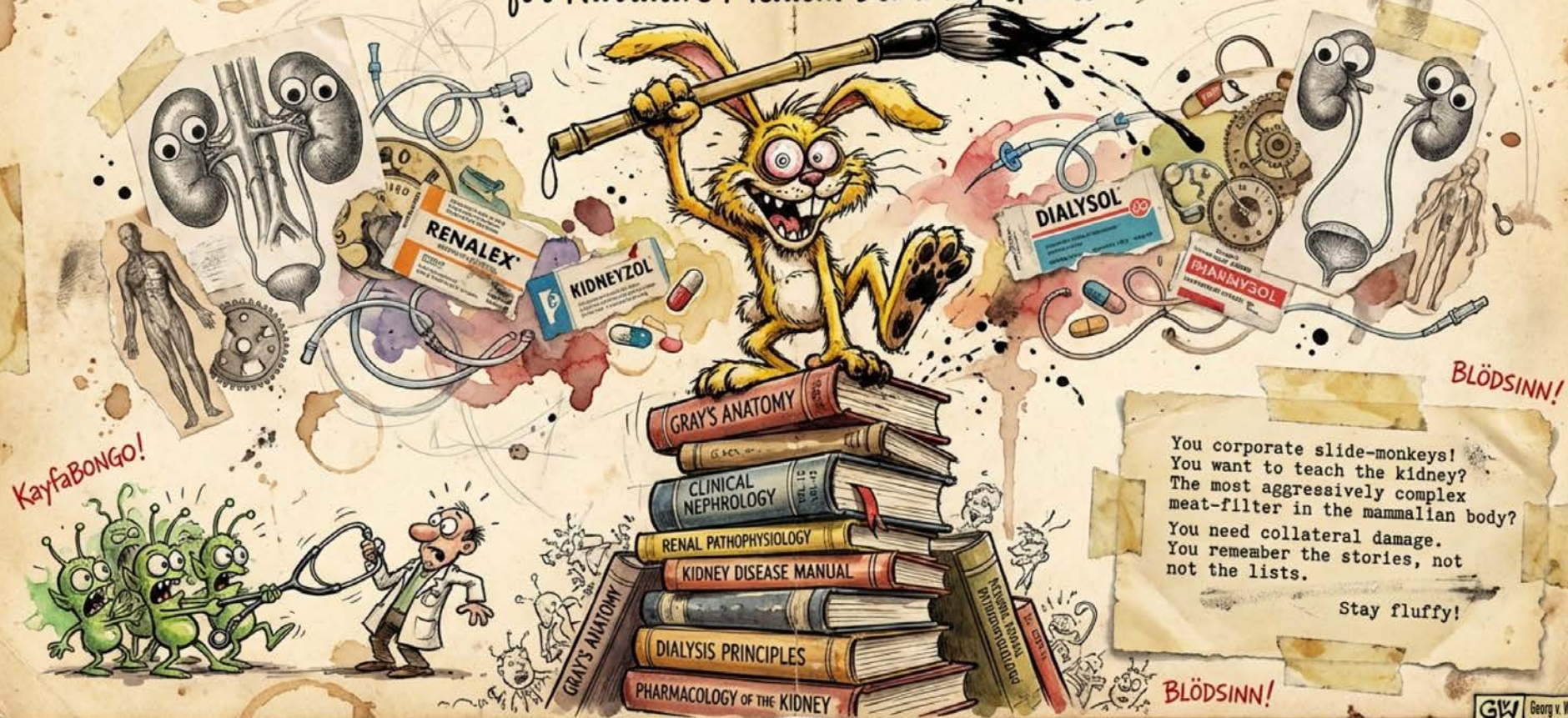


Georg von Westphalen

MedKayfab: Nephrology Edition

Uncle FrizzleBob's Krazy Kayfabizzaro Komik Kard Kult Klub
for Narrative Medical Learning Games

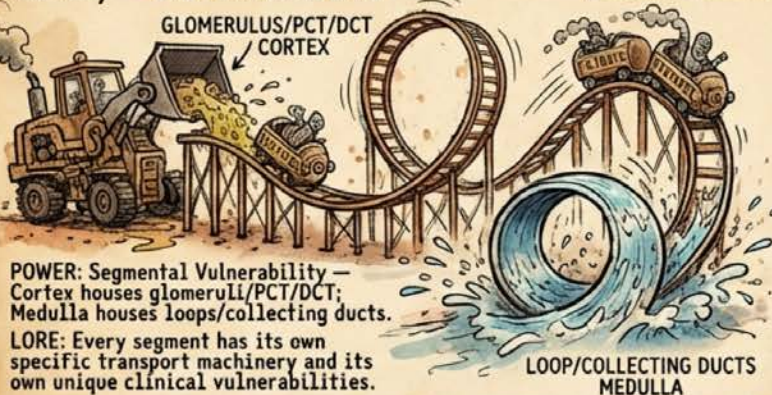


Nephro the Nephron

Anatomy — The Functional Unit

STAT Block
180 Litres / Day

GLOMERULUS/PCT/DCT
CORTEX



POWER: Segmental Vulnerability — Cortex houses glomeruli/PCT/DCT; Medulla houses loops/collecting ducts.
LORE: Every segment has its own specific transport machinery and its own unique clinical vulnerabilities.

KayfaBIZARRO!



Filtration Barry

Glomerular Barrier Structure

STAT Block
< 4nm Size Limit

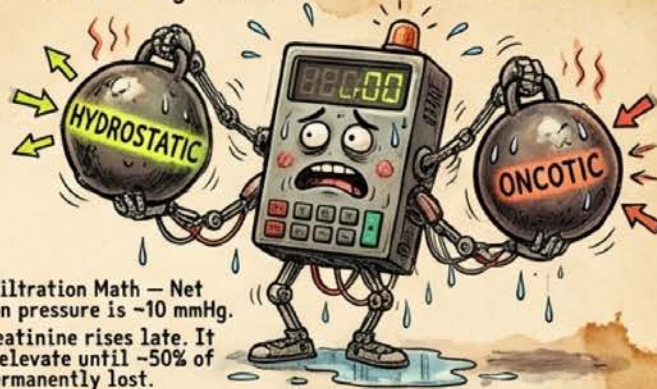


POWER: Charge & Size Repulsion — Defends against albumin leak.
LORE: Loss of heparan sulfate from the GBM causes selective albumin leak before structural size-barrier failure.

GFR Gary

Measurement & Starling Forces

STAT Block
~120 mL/min

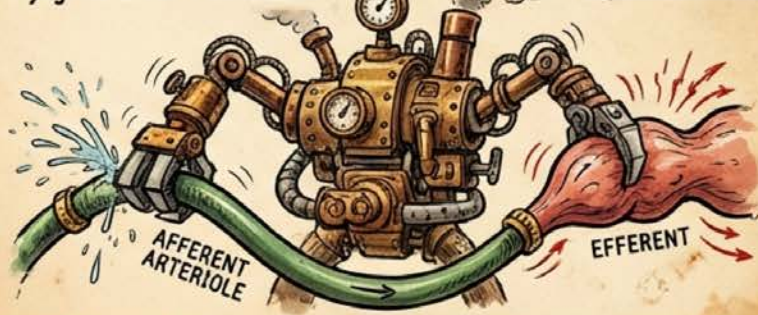


POWER: Filtration Math — Net filtration pressure is ~10 mmHg.
LORE: Creatinine rises late. It will not elevate until ~50% of GFR is permanently lost.

Auto the Autoregulation

Myogenic & TGF Mechanisms

STAT Block
MAP 75–160 mmHg

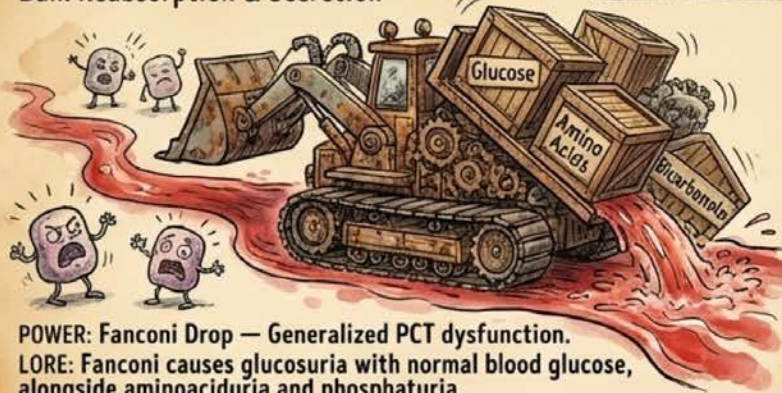


POWER: Pressure Defense — Tubuloglomerular feedback manages NaCl delivery.
LORE: ACE inhibitors dilate the efferent arteriole; NSAIDs constrict the afferent. Together, they drop GFR.

Tubulo the PCT

Bulk Reabsorption & Secretion

STAT Block
65% Na Retained



POWER: Fanconi Drop — Generalized PCT dysfunction.
LORE: Fanconi causes glucosuria with normal blood glucose, alongside aminoaciduria and phosphaturia.

HUMBUG!



Disco the DCT

NCC & Calcium Fine-Tuning

STAT Block
5% Na Load



POWER: Thiazide Shift — Blocks NCC.
LORE: Thiazides block NCC, causing volume depletion which triggers hypercalcaemia and hypocalciuria.

Henley the Loop

Countercurrent Multiplication

STAT Block
1200 mOsm/kg



POWER: Medullary Concentrating Engine.
LORE: Loop diuretics block NKCC2, completely abolishing the medullary gradient and causing massive diuresis.

Collect the Collecting Duct

Aldosterone & ADH Command

STAT Block
Urine pH < 5.5



POWER: Hormonal Final Say.
LORE: Liddle syndrome features an ENaC gain-of-function mutation that phenocopies aldosterone excess, but only responds to amiloride.

"Nephrotic Ned"

Protein Leak & Oedema

> 3.5g Protein/Day



POWER: Oncotic Collapse — Loss of plasma oncotic pressure.

LORE: Loss of urinary anticoagulants makes this a hypercoagulable state. Membranous nephropathy carries the highest renal vein thrombotic risk.

"IgA Ivan"

Berger's Disease

Most Common Primary GN



POWER: Mesangial Deposition Strike.

LORE: Haematuria occurs concurrently (synpharyngitic, 1-2 days) with the URTI, distinguishing it from post-streptococcal GN.

KayfaBINGO!

"Nephritic Neil"

Inflammation & Red Cell Casts

Protein < 3.5g/Day



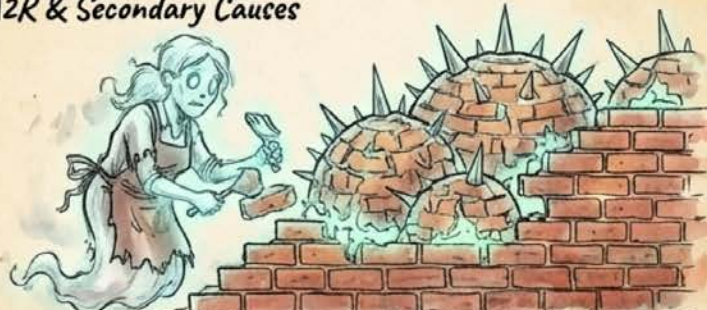
POWER: Inflammatory Blockade — Reduced GFR, oliguria, hypertension.

LORE: Red cell casts are formed in the tubules and are strictly pathognomonic of glomerular bleeding.

"Membranous Mary"

PLA2R & Secondary Causes

80% Anti-PLA2R



POWER: Spike & Dome Subepithelial Thickening.

LORE: Secondary causes account for 20%. You must screen for occult solid tumour malignancy in all adults over 60.

KayfaBINGO!



"Minimal Chang"

The Podocyte Enigma

STAT Block
>90% of Child Nephrotic



POWER: T-Cell Permeability Factor.

LORE: Dramatic, selective albuminuria with essentially normal light microscopy. Highly responsive to steroids.

What the FLUFF?!



"FSGS Frankie"

Sclerosis & Scarring

STAT Block
Non-Selective Proteinuria



POWER: Podocyte Collapse — Loss is irreversible.

LORE: Recurrence after renal transplantation in primary FSGS strongly indicates a circulating permeability factor (like suPAR).

"Lupus Nephritis Lucie"

ISN/RPS Classification

STAT Block
Class IV = Diffuse & Worst



POWER: Immune Complex War — 'Wire loop' subendothelial deposits.

LORE: The 'Full House' immunofluorescence pattern is virtually diagnostic of lupus nephritis.

"ANCA Vasculitis Vince"

GPA, MPA, EGPA

STAT Block
Pauci-Immune



POWER: Endothelial Arson — Crescent formation without immune deposits.

LORE: Presents as a pulmonary-renal syndrome: rapidly progressive GN combined with alveolar haemorrhage.

"Post-Strep Percy"

Immune Complex Nephritis

STAT Block
1-3 Weeks Latency



POWER: Alternative Pathway Consumption.

LORE: C3 is low but C4 is normal, revealing isolated alternative complement pathway activation.



KayfaBOGGLE!



"Alport the Hereditary"

Type IV Collagen Defect

STAT Block
80% X-Linked



POWER: Basement Membrane Lamellation.

LORE: Type IV collagen mutations cause progressive GBM splitting, leading to nephritis and sensorineural deafness.

"Acid-Base Alice"

Henderson-Hasselbalch Framework

STAT Block
pH 7.35 - 7.45



POWER: Respiratory & Renal Compensation.

LORE: Winter's Formula: Expected $pCO_2 = 1.5 \times HCO_3 + 8 \pm 2$. Use it to check if respiratory compensation is appropriate.



"Anion Gap Andy"

High AG Metabolic Acidosis

STAT Block
Normal AG = 8-12



POWER: MUDPILES Protocol — Methanol, Uraemia, DKA, Propylene Glycol, Isoniazid, Lactic Acid, Ethylene Glycol, Salicylates.

LORE: Metformin causes Type B lactic acidosis (without tissue hypoxia), driving a pure high anion gap metabolic acidosis.

Stay fluffy!

Normal AG Normo

RTAs & GI Loss

STAT Block
Hyperchloraemic
Acidosis



POWER: Bicarbonate Wasting / H^+ Retention.

LORE: Type I distal RTA fails to secrete H^+ , meaning the urine can never be acidified below pH 5.5 despite systemic acidosis.

22

Metabolic Alkalosis Maggie

Generation & Maintenance

STAT Block
Urine Cl
< or > 20



POWER: The Maintenance Lock.

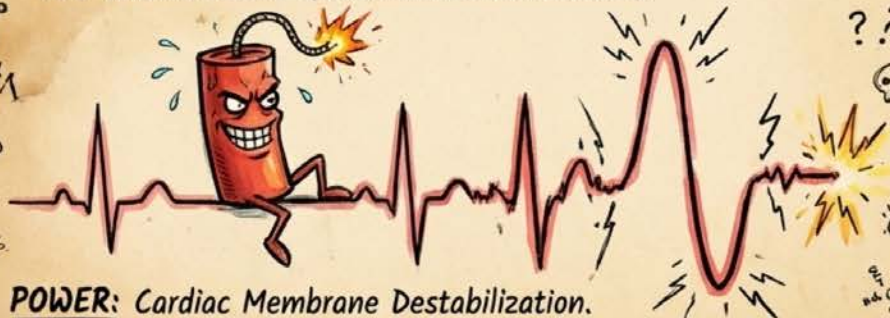
LORE: The kidney can excrete any HCO_3 load, unless locked into maintenance by hypovolaemia or hypokalaemia.

23

Potassium Perry

The Widowmaker (Internal/External Balance)

STAT Block
98% Intracellular



POWER: Cardiac Membrane Destabilization.

LORE: In hyperkalaemic arrest, give Calcium Gluconate first. It does not lower K^+ ; it stabilizes the myocardium to prevent VF.

24

Sodium Stella

Osmolality & Volume

STAT Block
Na < 135 or > 145



POWER: Tonicity Shift.

LORE: Correct hypernatraemia slowly (<10 mmol/L per 24h). Rapid correction floods the adapted brain, causing cerebral edema.

25

Magnesium Magda

The Forgotten Electrolyte

STAT Block
1% Extracellular



POWER: The Refractory Block.

LORE: Hypomagnesaemia impairs ROMK function. Hypokalaemia will be completely refractory to K⁺ replacement until Mg is fixed.

27

RTA Rita

Full Clinical Classification

STAT Block
3 Distinct
Tubular Defects



POWER: The Acid Trap Matrix.

LORE: Type 4 RTA (hyporenal aldosteronism) is the only RTA characterized by high serum potassium.

26

Phosphate Phil

FGF-23 & PTH Axis

STAT Block
85% in Bone



POWER: Cellular Energy Depletion.

LORE: Refeeding syndrome drives phosphate intracellularly, causing severe ATP depletion, haemolysis, and respiratory failure.

28

Uric Acid Ulric

Gout & Renal Handling

STAT Block
70% Renally
Excreted



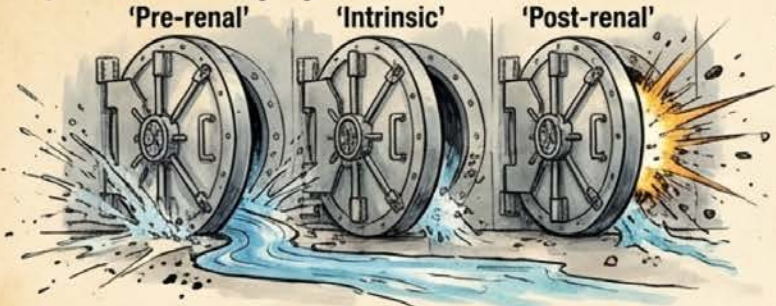
POWER: Inflammasome Trigger — MSU crystals activate macrophages.

LORE: Never initiate allopurinol (urate-lowering therapy) during an acute gout attack; it mobilizes stores and worsens the flare.

29

AKI the Acute Definition & Staging

STAT Block:
FENa < 1% = Pre-Renal



POWER: The Staging Triad — Creatinine rise, or urine output drop.
LORE: Pre-renal AKI presents with urine Na < 20 and a concentrated urine osmolality > 500 mOsm/kg.

30

ATN the Tubule Destroyer Ischaemia & Toxins

STAT Block:
Muddy Brown Casts

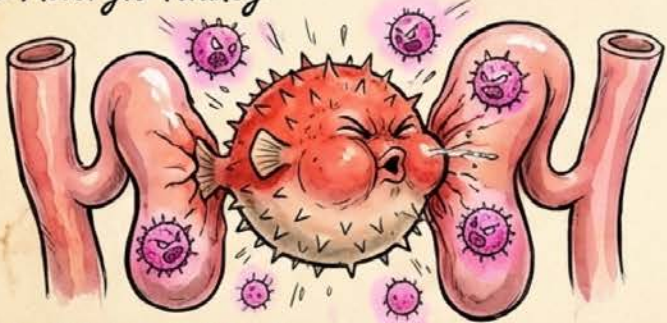


POWER: Intraluminal Obstruction.
LORE: Rhabdomyolysis causes haem-pigment ATN; it requires aggressive IV isotonic saline to prevent tubular obstruction.

31

AIN the Interstitial The Allergic Kidney

STAT Block:
Classic Triad in only 10%



POWER: T-Cell Infiltration & Tubulitis.
LORE: PPI-associated AIN often lacks the classic fever/rash and presents insidiously as unexplained CKD

32

Hepatorenal Hugo Functional Renal Failure

STAT Block:
Visceral Vasodilation



POWER: Maximal Renal Vasoconstriction.
LORE: Kidneys from HRS patients function perfectly if transplanted. The failure is entirely haemodynamic, driven by cirrhosis.

33

CKD the Chronic

Staging & Systemic Consequences

GFR < 60 for > 3 months



POWER: Systemic Dominoes.

LORE: Cardiovascular disease is the leading cause of death in CKD. Most patients die of heart disease before reaching dialysis.

34

Dialysis Danny

Haemodialysis Mechanics

Target Kt/V > 1.2



POWER: Diffusion & Ultrafiltration.

LORE: Arteriovenous fistulas (AVF) are preferred over central lines because they carry the lowest risk of infection and thrombosis.

35

Peritoneal Pearl

At-Home Fluid Exchanges

1.5% - 4.25% Glucose



POWER: Osmotic Pull — Glucose drives ultrafiltration.

LORE: Peritoneal dialysis preserves residual renal function much better than intermittent haemodialysis.

36

Transplant Tina

Immunosuppression

Calcineurin Inhibitor Toxicity



POWER: Triple Therapy Defense.

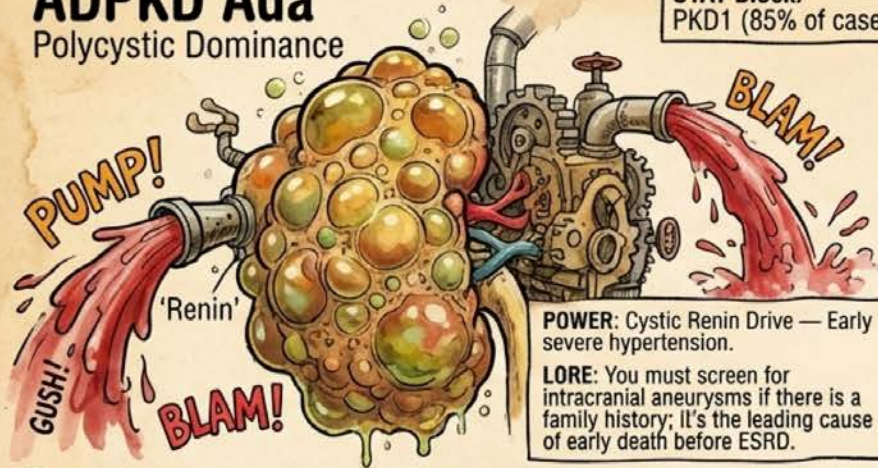
LORE: Chronic allograft nephropathy (fibrosis/atrophy) remains the leading cause of late graft loss, ironically worsened by CNI nephrotoxicity.

GW Georg v. Westphalen

37

ADPKD Ada

Polycystic Dominance



STAT Block:
PKD1 (85% of cases)

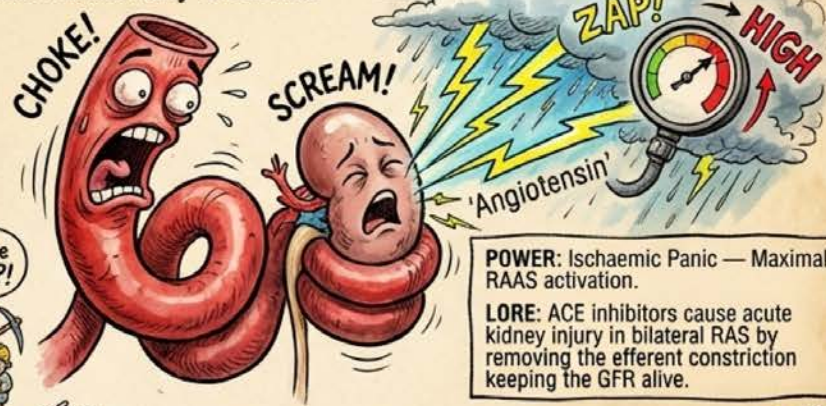
POWER: Cystic Renin Drive — Early severe hypertension.

LORE: You must screen for intracranial aneurysms if there is a family history; it's the leading cause of early death before ESRD.

38

Renovascular Rex

Flash Pulmonary Oedema



STAT Block:
Bilateral RAS

POWER: Ischaemic Panic — Maximal RAAS activation.

LORE: ACE inhibitors cause acute kidney injury in bilateral RAS by removing the efferent constriction keeping the GFR alive.

39

Nephrolithiasis Nick

Stone Typologies



STAT Block:
70% Calcium Oxalate

POWER: Supersaturation Crystallization.

LORE: Never restrict dietary calcium for calcium stones; doing so increases free gut oxalate absorption, worsening the stones.



40

Proteinuria Petra

The Distress Signal



STAT Block:
ACR < 3 mg/mm (Normal)

POWER: The Dipstick Limit.

LORE: Standard urine dipsticks only detect albumin. They will read negative for the overflow tubular Bence Jones proteins in myeloma.

41 **ACEi Amy**
The Nephroprotector

STAT Block
Acceptable eGFR
fall $\leq 25\%$



POWER: Intraglomerular Decompression.
LORE: Dual RAAS blockade (ACEi + ARB) provides no additional renal benefit but massively increases the risk of AKI and hyperkalaemia.

42 **Diuretics Dmitri**
The Volume Controller

STAT Block
Furosemide
ceiling 1000mg



POWER: Sequential Nephron Blockade
LORE: Severe nephrotic patients need IV loop diuretics because massive gut wall oedema prevents oral drug absorption.

KayfaBONGO!



43 **Erythropoietin Ernie**
Anaemia Driver

STAT Block
Target Hb
100–120 g/L



POWER: Hypoxia Sensor & Red Cell Proliferation.
LORE: Pushing Hb targets higher than 120 g/L in CKD increases the risk of stroke without providing any mortality benefit.

44 **Phosphate Binder Percy**
The Gut's First Line

STAT Block
Prevents
CKD-MBD



POWER: Intestinal Chelation.
LORE: Binders must be taken exactly with meals. Calcium-based binders risk worsening vascular calcification in severe CKD.



(45)

Calcimimetics Carla

CaSR Trickery

STAT Block:
Suppresses PTH



POWER: Allosteric Modulation.

LORE: Preferred for secondary hyperparathyroidism in dialysis because it lowers PTH without raising calcium, unlike active vitamin D.

(46)

Contrast Nephropathy Connie

Iodinated Destruction

STAT Block:
Cr rise >25% in 48h



POWER: Medullary Ischaemia + Direct Toxicity.

LORE: Metformin is not nephrotoxic itself, but must be held 48 hours prior to contrast to prevent fatal lactic acidosis if AKI occurs.

(47)

TMA Tamara

HUS & TTP

STAT Block:
Schistocytes



POWER: Microvascular Shearing & Platelet Consumption.

LORE: Antibiotics are absolutely contraindicated in STEC-HUS; killing the bacteria triggers a massive release of Shiga toxin.

(48)

Myeloma Kidney Mick

Cast Nephropathy

STAT Block:
Polychrome Fractured Casts



POWER: Tubular Overload & Obstruction.

LORE: Myeloma cast nephropathy causes severe AKI, which must be distinguished from AL amyloidosis which causes nephrotic syndrome.

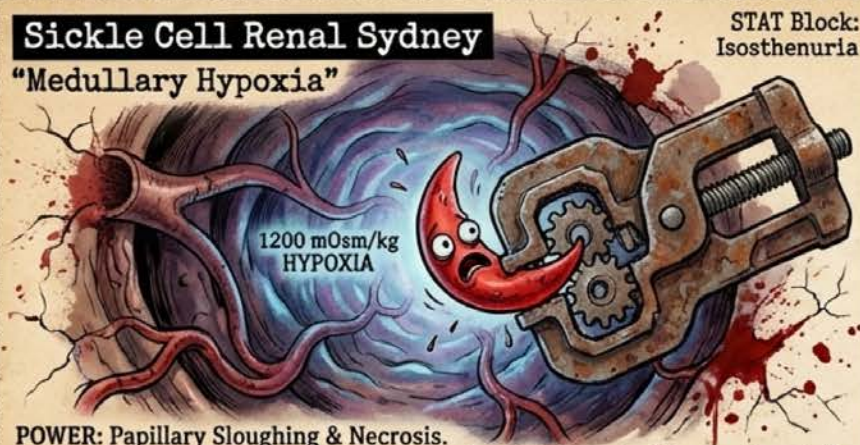
HUMBUG!



Sickle Cell Renal Sydney

"Medullary Hypoxia"

STAT Block:
Isosthenuria



POWER: Papillary Sloughing & Necrosis.

LORE: The extreme 1200 mOsm/kg hypoxic environment of the renal medulla causes RBC sickling and damage even in sickle cell trait.

Lupus Anticoagulant Lena

"The APTT Paradox"

STAT Block:
Antiphospholipid Syndrome



POWER: In-Vitro Paradox / In-Vivo Thrombosis.

LORE: 'Lupus Anticoagulant' is a misnomer. It prolongs the APTT in laboratory tests but causes catastrophic micro-thrombosis clinically.

Kidney Biopsy Betty

"Light, IF, EM"

20-25 Glomeruli Needed



POWER: Tri-Partite Processing.

LORE: Every core must be processed for Light Microscopy, Immunofluorescence, and Electron Microscopy simultaneously to diagnose glomerular disease.

Urinalysis Ursula

"The Nephrologist's Window"

≥ 1 RBC/HPF = Haematuria



POWER: Non-Invasive Diagnostics.

LORE: Broad waxy casts indicate advanced CKD; the tubules have dilated to compensate for mass nephron dropout, creating wide, slow-flowing casts.

53 Electrolyte Emergencies Eddie

Priority Management

$K > 6.5 = \text{Immediate Danger}$



POWER: Cardiac Hierarchy Protocol.

LORE: Correct hyponatraemia slowly (max 8-10 mmol/L/24h). Fast correction destroys the brain via osmotic demyelination syndrome.

55 The Nephrology Attending

Master Review Synthesis

100% Clinical Integration



POWER: Mechanistic Justification.

LORE: Map every clinical diagnosis to its cellular mechanism. Justify every treatment decision with an underlying physiological principle. No approximations.

54 RAAS Pharmacology Renata

The Blockade Ladder

Non-Steroidal MRA (Finerenone)



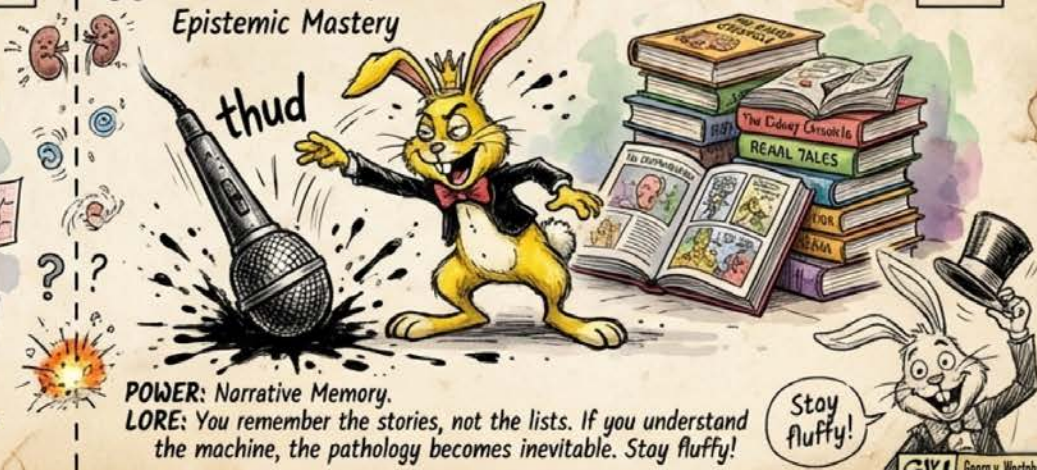
POWER: Pleiotropic Protection.

LORE: Finerenone is a non-steroidal MRA preferred in diabetic CKD; it reduces fibrosis independently of its haemodynamic effects.

56 FrizzleBob's Golden Rule

Epistemic Mastery

∞



POWER: Narrative Memory.

LORE: You remember the stories, not the lists. If you understand the machine, the pathology becomes inevitable. Stay fluffy!